

Lev

Lev is a human-centered agent MVP for a live document workspace.

It opens a local project folder, lets the user write in a document, observes the document changes, infers the user's current intent, and shows quiet assistance in a read-only side panel. Lev intentionally has no chat input in the main UI.

Product Demo

Lev keeps the user's main action in the document editor and surfaces quiet, read-only assistance beside it.

The screenshot displays the Lev application interface. At the top left is the Lev logo (a green square with a white 'L') and the text 'Lev Human-centered document workspace'. To the right is a 'Document' dropdown menu showing 'README.md' and a 'Pause' button. The main area is split into two panels. The left panel, titled 'README.md' and 'Saved', contains the following text:

```
# Lev

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observes the document changes, infers the user's current intent, and
shows quiet assistance in a read-only side panel. Lev intentionally
has no chat input in the main UI. 你好

## Product Principle

Lev is not trying to replace the user's subject action. In this MVP,
the subject action is writing and thinking inside a document. Lev
should prepare context,
surface useful references, expose friction, and suggest small next
moves without
taking over the document.

The shortest design rule is:

> Do not complete the user's meaningful action. Arrange the workspace
so the
> user can complete it more smoothly.
```

The right panel, titled 'Lev assistance' and 'Local', contains the following text:

当前推断: product_design

我看到你正在编辑 `README.md`。这一版先不替你写正文，只把最近变化和光标附近内容整理出来，方便你继续自己的判断。

最近变化:

--- previous

+++ current

@@ -2,7 +2,7 @@

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What The MVP Does

- Opens a folder as a Lev workspace.
- Lists .md and .txt documents in that workspace.
- Provides a two-pane workbench:
 - left: editable document
 - right: read-only Lev assistance
- Autosaves the active document.
- Sends document diffs, cursor position, and selection offsets to the agent.
- Reuses the existing agent loop and search/page-reading tools.
- Falls back to local diff/cursor-context assistance if the model or API key is unavailable.

Project Structure

client.py	AI Builders Space client
tools.py	web_search and read_page tools
main.py	original one-shot agent loop CLI
lev/	
assist.py	document-change -> prompt -> assistance
models.py	dataclass request/response models
server.py	no-dependency local HTTP server
workspace.py	local document listing/read/write
web/	
index.html	Lev workbench shell
app.js	autosave, cursor tracking, assistance requests
styles.css	two-pane UI
docs/	
lev-mvp.md	MVP behavior and evaluation
editor-decision.md	editor/fork decision

Setup

uv sync

Create .env in this folder:

BUILDER_API_KEY=your_api_key_here

client.py reads this local .env file.

Run Lev

Open the current repository as the workspace:

```
uv run python -m lev.server .
```

Then open:

```
http://127.0.0.1:8787
```

Open another folder as the workspace:

```
uv run python -m lev.server /path/to/project
```

Change the port if needed:

```
uv run python -m lev.server . --port 8790
```

Keep The Original Agent Loop

The original one-shot CLI still works:

```
uv run python main.py "Search for the latest Python release."
```

Lev uses that loop as a lower-level capability. The new HC layer changes the entrypoint from "user prompt" to "human activity inside a document".

Editor Direction

The MVP uses a native `<textarea>` behind a small frontend boundary. This keeps the first version dependency-free and easy to run.

For the next step, use CodeMirror 6 rather than forking a full editor product. CodeMirror is much lighter than Monaco, has first-class document state, selection, cursor, transactions, decorations, and programmatic scrolling, and fits Lev's document-workbench shape. Monaco is excellent for code-IDE behavior but is a heavier starting point for this writing/research MVP.

See <docs/editor-decision.md>.